

More examples = -

① For any vector $v = (v_1, v_2) \in \mathbb{R}^2$, define $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ by =

$$T(v_1, v_2) = (v_1 - v_2, v_1 + 2v_2)$$

(a) Find the image of $v = (-1, 2)$

(b) Find the image of $v = (0, 0)$

(c) Find the pre image of $w = (-1, 11)$

Sol

(a) For $v = (-1, 2)$, we have

$$T(-1, 2) = (-1 - 2, -1 + 2(2)) = (-3, 3)$$

(b) If $v = (0, 0) \Rightarrow T(0, 0) = (0 - 0, 0 + 2(0)) = (0, 0)$

(c) $T(v) = (v_1 - v_2, v_1 + 2v_2) = (-1, 11)$

$$v_1 - v_2 = -1$$

$$v_1 + 2v_2 = 11$$

; then solve the system

This system of equations has the unique solution $v_1 = 3$ and $v_2 = 4$.

So, $(3, 4)$ is the pre image of $(-1, 11) \in \mathbb{R}^2$.

② Show that the function in ①, is a linear transformation

$$T(v_1, v_2) = (v_1 - v_2, v_1 + 2v_2)$$

$$\text{let } u = (u_1, u_2) \quad , \quad v = (v_1, v_2) \in \mathbb{R}^2$$

$$u+v = (u_1+v_1, u_2+v_2)$$

$$\Rightarrow T(u+v) = ((u_1+v_1)-(u_2+v_2), (u_1+v_1)+2(u_2+v_2))$$

$$= (u_1+v_1-u_2-v_2, u_1+v_1+2u_2+2v_2)$$

$$= ((u_1-u_2)+(v_1-v_2), (u_1+2u_2)+(v_1+2v_2))$$

$$= (u_1-u_2, u_1+2u_2) + (v_1-v_2, v_1+2v_2)$$

$$= T(u) + T(v)$$

$$(ii) \quad cu = c(u_1, u_2) = (cu_1, cu_2)$$

$$T(cu) = T(cu_1, cu_2) = (cu_1 - cu_2, cu_1 + 2cu_2)$$

$$= c(u_1 - u_2, u_1 + 2u_2) = cT(u) \neq$$

So T is linear transformation.

Some functions are not linear:

$$(i) \quad f(x) = \sin x$$

$$(ii) \quad f(x) = x+1$$

$$(iii) \quad f(x) = x^2$$